Housing First or Treatment First?

Model, calibration, validation, and mechanisms

Draft — August 14, 2026

Abstract

This note documents the rebuilt model, its two-stage calibration, the untargeted moments it is validated against, and the mechanism decomposition that replaces the appendix's static model. It is written to be read as a draft of Sections 2–5 of the revision. Numerical results from the calibration and the mechanism ladder are marked where they are still pending; the specifications, targets and sources are final.

Contents

1 The model **2**

1.1 Environment . . . 2
1.2 Preferences . . . 3
1.3 Labour supply . . . 4
1.4 Drug use, addiction and treatment . . . 5
1.5 Housing, vouchers, and the two policies . . . 5
1.6 Serious mental illness . . . 5
1.7 Government and equilibrium . . . 6

2 Calibration **6**

2.1 First stage: the stationary equilibrium . . . 6
2.1.1 Parameters set outside the model . . . 6
2.1.2 Internally calibrated parameters . . . 7
2.2 Second stage: short-run experimental moments . . . 9
2.2.1 The housing-access elasticity . . . 9
2.2.2 The drug-price participation elasticity . . . 10
2.2.3 The treatment effect and the value of a statistical life . . . 10

3 Validation **11**

3.1 Distributional moments . . . 11

Table 1: State space: 12,960 points per age

State	Symbol	Points	Notes
Age	j	60	real age 26–85, retirement at 66
Financial assets	$a \geq 0$	60	cubic spacing, dense near zero; no borrowing
Labour productivity	y	9	Rouwenhorst discretisation of an AR(1) in logs, $\rho = 0.90$, $\sigma = 0.20$
Employment	e	3	employed; unemployed with UI; unemployed without UI
SUD status	d	2	clean; substance use disorder
Serious mental illness	m	2	absent; present
Housing voucher	η	2	not held; held

3.2 Policy responses 11

4 Mechanisms: shutting channels and recalibrating b 12

4.1 Why this replaces the static appendix 12

4.2 The design 12

4.3 The arms 13

4.4 What is reported 13

4.5 What the exercise is expected to show, and how it could fail 13

4.6 Where the externalities come from 14

5 Status 15

1 The model

1.1 Environment

Time is discrete and a period is one year. The economy is populated by overlapping generations of households indexed by age $j = 1, \ldots, J$, corresponding to real ages 26 to 85, with retirement at real age 66 ($j_R = 41$). Households face age-, health- and behaviour-dependent mortality; the assets of the deceased are redistributed to survivors as a uniform lump-sum bequest.

The individual state is

$$s = \big(j,\ a,\ y,\ e,\ d,\ m,\ \eta\big),$$

with $a \geq 0$ financial assets, y labour productivity, e employment and unemployment-insurance status, $d \in \{0, 1\}$ substance use disorder (SUD), $m \in \{0, 1\}$ serious mental illness (SMI), and $\eta \in \{0, 1\}$ possession of a housing voucher. Table 1 gives the discretisation.

Two features of the state space are deliberate and are worth stating because they are what make the model solvable at this size.

Housing entered with is not a state. Survival, employment transitions and the SUD recovery hazard all condition on the *chosen* housing h', and renters face no adjustment cost, so the unit a household arrives in is payoff-irrelevant. Carrying it would replicate the entire problem N_h times. Removing it leaves every aggregate unchanged and cuts the solve time by a factor of 6.6.

Three employment states, not four. Unemployment insurance is drawn on *entry* into unemployment with probability $\pi_{UI} = 0.30$ and lasts one year. A first-year non-recipient and a household unemployed for two or more years have identical income (zero) and identical continuation values, so those two states merge *exactly*. This is an equivalence, not an approximation, and it answers the Associate Editor's question A2 about whether modelling UI expiration is important: it is, but it costs one state rather than two.

1.2 Preferences

Period utility over consumption c, housing services h, drug consumption v and hours n is

$$u(c,h,v,n;d,m,h) = \frac{\left[c^{\alpha}h^{\beta}(v+b_{dmh})^{\gamma}\right]^{1-\sigma}}{1-\sigma} + \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} + \aleph - \kappa_x\big(H + \omega_x H_D\big), \qquad (1)$$

with

$$\alpha = (1-\theta_h)(1-\theta_v), \qquad \beta = \theta_h(1-\theta_v), \qquad \gamma = \theta_v.$$

Cobb–Douglas weights do not depend on SUD status. This is the change requested by Referee 2 (C5), Referee 3 (M2a) and the AE (A8). In the previous specification the taste weight on drugs rose with d, so reducing SUD mechanically raised the utility derived from c and h, making addiction-reduction and housing expansion complements "even absent incentive effects". Holding α, β, γ fixed removes that channel entirely.

Only the Stone–Geary shifter varies. b_{dmh} depends on SUD status, mental illness and *chosen housing*:

$$b_{dmh} = b_d \cdot \underbrace{\rho_m^{\mathbf{1}\{m=1\}}}_{\text{self-medication}} \cdot \underbrace{\rho_H^{\mathbf{1}\{h=\text{homeless}\}}}_{\text{housing access}}, \qquad \rho_m, \rho_H \in (0,1]. \qquad (2)$$

A lower b raises the marginal utility of the first units of drug use, so poor mental health and homelessness both push a household towards initiating. Three remarks.

First, the AE's suggestion (A8) was to keep the Cobb–Douglas weights invariant and vary b instead. That is what (2) does. A *single* b cannot match both the participation moment and the drug-expenditure share: a b large enough to deter clean households pushes the unconstrained interior choice v^* below zero, which drives drug spending by households with SUD to the corner. Two values, b_0 for clean and b_1 for SUD, separate the two moments — b_0 pins participation and b_1 keeps the choice interior so that θ_v pins the expenditure share.

Second, $\rho_m < 1$ is the self-medication channel Referee 1 asks about in comment 1(a).

Third, and most consequentially, ρ_H **is the only parameter in the model that governs the elasticity of drug use with respect to housing access**. Referee 2 calls this elasticity the

crux on which all the welfare results rest; Referee 3 and the AE (A13) say the same independently. Without ρ_H the moment is not merely untargeted, it is *untargetable*: housing a household raises its resources, drugs are a normal good, and the model would predict that housing *raises* use — the opposite sign to Milby et al. (2005). Section 2.2 calibrates ρ_H to that trial.

The value of life. $\aleph$ is the flow value of being alive. It is what makes mortality risk deter drug use, and it therefore carries the paper's central mechanism; Section 4 shows exactly how much work it does.

The externality. H is the aggregate homelessness rate and H_D the share of the population both homeless and with SUD. Referee 2's wording is specific — "a large homeless population, *especially if many are addicted to drugs*, creates a negative externality beyond the fiscal burden" — so the disamenity is weighted by composition through ω_x, not by the headcount alone. Households take H and H_D as given, which is what makes this a spillover rather than a private cost, but both are equilibrium objects, so policy moves them and the movement feeds back through the value of life into drug decisions. This is precisely the property the loss-of-consortium adjustment κ_{cons} lacked, and which Referee 1 objected to in comment 3(a). The baseline sets $\kappa_x = 0$; see Section 4.6.

1.3 Labour supply

Hours are continuous and are pinned by the intratemporal first-order condition,

$$u_c(c) \cdot \frac{\partial R(n)}{\partial n} = \omega_\ell (1-n)^{-\sigma_\ell}, \tag{3}$$

where $R(n)$ denotes resources net of saving. Two features of (3) matter economically, not just numerically.

The residual is not monotone. The food-stamp floor is clawed back at a rate of one and the housing voucher at the tenant contribution rate $\kappa_r = 0.30$, so $\partial R/\partial n$ *jumps up* at the hours where each transfer is exhausted, and the left-hand side of (3) can cross the right-hand side more than once. This is the welfare trap: below the food-stamp exhaustion point the return to an extra hour is close to zero, so "work a little and keep the transfer" and "work properly" are genuinely distinct local optima and the household must compare them. The solution algorithm splits $[0, 1]$ at the kinks and solves each piece exactly rather than bracketing once, which would silently select whichever optimum happened to be bracketed.

Voucher eligibility is a step in the continuation value. Income eligibility is lost discretely at 50% of median income. That does not kink the budget, so (3) is continuous across it; it shifts the continuation value, so the piecewise optimum is clamped to each side of the eligibility threshold and both are evaluated.

Referee 3 (m1) wanted labour supply retained and its role made visible; the AE (A3) questioned whether it was needed at all. Making it continuous and exact resolves the tension in Referee 3's favour at a cost of a factor of three in solve time.

1.4 Drug use, addiction and treatment

A household chooses whether to use ($v > 0$) or abstain ($v = 0$), and how much, subject to the price p_v. A clean household that uses acquires SUD next period with probability π_d, *independent of how much it consumes*. There is no light/heavy threshold and no consumption-dependent hazard; the previous specification's $\bar{v}$ and the 6.48 heavy-to-light ratio are gone. This is a simplification, and it removes two of the parameters Referee 3 (M2b) asked to have justified.

A household with SUD recovers with hazard

$$\pi_r(h, \tau, m) = \bar{\pi}_r \cdot \mu_H^{\mathbf{1}\{h=\text{homeless}\}} \cdot \mu_m^{\mathbf{1}\{m=1\}} \cdot g^{\tau}, \tag{4}$$

where $\tau \in \{0, 1\}$ is a discrete **treatment take-up** decision carrying a resource cost, and $g > 1$ is the treatment effect on the recovery hazard. Referee 1's comment 1(b) and the AE's A10 are the same point and it is the most damning gap in the submitted paper: the policy is called Treatment First and there is no treatment in the model. There is now.

1.5 Housing, vouchers, and the two policies

Rent is a fixed price; there is no housing-market general equilibrium. This follows the AE's A5 — prices and rents cleared two markets whose PE and GE outcomes were nearly identical. Homelessness is the lowest point of a four-point housing grid with a calibrated service flow, replacing the seven-point grid with four uncalibrated values that Referee 2 (D1) objected to.

Voucher rules are institutional and are not free parameters: income eligibility below 50% of median income, a 30% tenant contribution, and a lottery probability calibrated to HUD take-up among the eligible. Vouchers pay only while renting.

The two policies differ in one rule:

	Housing First	Treatment First
Voucher conditional on abstinence	no	yes
Detection probability	—	$p_{\text{det}} \leq 1$
False-positive rate	—	$p_{\text{fp}} \geq 0$

Referee 1 (4a, 4b) and the AE (A11) both note that perfect observation and enforcement are not credible. p_{det} and p_{fp} are experiment parameters reported as sensitivity, not calibrated, because that is exactly the point.

1.6 Serious mental illness

m is an exogenous persistent binary state representing **serious** mental illness, not any mental illness. The distinction is not cosmetic: AMI covers roughly 23% of US adults and includes mild anxiety and depression, whereas SMI is about 5.5% and is defined by substantial functional impairment. It is SMI that drives chronic homelessness and disability receipt, and a state that halves productivity cannot sensibly cover a quarter of the population.

The specification follows İmrohoroğlu and Zhao (2022) structurally as well as numerically. In their model labour-market shocks generate *short* homeless spells while health shocks generate

chronic homelessness, and expanded means-tested disability income is the effective policy for the chronic group. Homelessness driven purely by income risk is transitory; the persistent tail requires a persistent state. SMI affects six objects: productivity conditional on working, job finding and separation, mortality, the SUD recovery hazard, the Stone–Geary shifter through ρ_m in (2), and eligibility for disability income when not employed. Onset is *derived* from the target prevalence and the recovery rate, so calibrating persistence never disturbs prevalence.

The productivity penalty is also what generates *employed* households poor enough to be priced out of housing — Referee 1's comment 2(c)(i), which an unemployment effect alone cannot deliver — and the persistence is what generates the older homeless population of comment 2(c)(ii).

1.7 Government and equilibrium

A stationary equilibrium is a set of prices and aggregates such that households optimise, the government budget balances, the drug market clears, bequests equal the assets of the deceased, and the externality aggregates are consistent:

Object	Determined by
τ_ℓ	government budget balance, closed form given the distribution
p_v	drug-market clearing, with production cost $C(V) = \frac{\psi_1}{\psi_2} V^{\psi_2}$, so the supply elasticity is $1/(\psi_2 - 1)$
Bequests	assets of the deceased, redistributed uniformly
H, H_D	fixed point, since they enter period utility

Making p_v an equilibrium object answers Referee 1 (1c), Referee 3 (M2d) and the AE (A5). Referee 3 worried that endogenising the price "requires suppressing this normalization". It does not: ψ_1 is normalised so that $p_v = 1$ at the benchmark, which leaves the existing normalisation intact and means only counterfactuals move the price.

Bequests are uniform rather than type-dependent, following the AE's A7.

2 Calibration

The calibration has two stages, and they are different exercises. The first stage matches a cross-section of the stationary equilibrium. The second stage matches short-run *responses* measured in experiments, which cannot be computed from a comparison of stationary equilibria.

2.1 First stage: the stationary equilibrium

2.1.1 Parameters set outside the model

Table 2: Externally set parameters

Parameter	Value	Source
σ	2.0	standard
σ_ℓ	3.0	Frisch elasticity of 0.5, Chetty (2012)
ξ (tax progressivity)	0.85	Heathcote, Storesletten and Violante (2017)
r	0.03	Kaplan, Mitman and Violante (2020)
κ_V (eligibility)	0.50× median	HUD programme rule
κ_r (tenant share)	0.30	HUD programme rule
π_{UI}	0.30	insured unemployment / total unemployment, FRED 2014–24
UI replacement rate	0.50	standard US
π_{ue} (job finding)	0.55	Krueger, Cramer and Cho (2014)
$\kappa_{ue,H}$	0.80	Desmond and Gershenson (2016)
$\kappa_{\phi,H}$	1.60	Meyer et al. (2025): 60% excess mortality when unhoused
$\kappa_{\phi,m}$	2.50	standardised mortality ratio for SMI of 2–3
SMI prevalence	5.5%	NSDUH 2023, adults
SMI recovery hazard	0.067/yr	İmrohoroğlu–Zhao's own moment: about half improve enough to work within ten years
χ_D	0.212	CEA (2017), roughly $30,000 per person-year
χ_H	0.252	National Alliance to End Homelessness; reported as sensitivity, since R2 and R3 both judge it high
$\rho_{\text{pen},d}$ (SUD wage penalty)	1.00	no direct wage effect in the baseline; see below

On the last row: the AE (A20) asks whether SUD has direct productivity effects. The robust findings are on the *employment* margin — roughly 50% more unscheduled absence and 44% higher turnover, which the model captures through $\kappa_{ue,d}$ — while estimates of the wage penalty conditional on working are contested, with parts of the older literature finding positive effects for casual use. The baseline therefore imposes no direct wage effect and says so, rather than inventing a number.

2.1.2 Internally calibrated parameters

All parameters are calibrated jointly, but the parameter-to-moment map is close to triangular: each parameter dominates one moment. Table 3 states the assignment. In the code the sign of $\partial M/\partial\theta$ is declared explicitly for each pair, so a mistaken sign shows up immediately as a moment walking away from its target rather than as a silent failure.

Table 3: Internally calibrated parameters and their moments

Parameter	Moment	Target	Source
b_0	drug participation	5.13%	NSDUH 2024, ages 26+
π_d	SUD rate	3.68%	NSDUH 2024, ages 26+
h_{homeless}	homelessness rate	0.30%	HUD point-in-time
π_{eu}	unemployment rate	4.90%	BLS
$\kappa_{ue,d}$	unemployment rate given SUD	26.90%	NSDUH
μ_H	drug use disorder share of the homeless	21.70%	Gutwinski et al. (2021)
π_V	voucher recipients / income-eligible	26.60%	HUD
θ_v	drug spending / income given SUD	44.00%	existing moment
θ_h	aggregate housing cost / income	21.20%	SCF 2022, all households
p_{rent}	share with housing cost above 50% of income	17.49%	SCF 2022, all households
$\aleph$	SUD income / average income	68.70%	NSDUH 2024
ζ_{DI}	SMI share of the homeless	25.00%	Gutwinski et al. (2021)
$\kappa_{ue,m}$	unemployment rate given SMI	7.10%	NSDUH Table 6.5B, by Bayes
$\rho_{\text{pen},m}$	homeless income / average	19.10%	Meyer et al., linked administrative data
β	**wealth / GDP**	**3.00**	standard US capital–output ratio
$\underline{c}$	food stamps / labour income	0.42%	SNAP outlays, FY2023
ρ_m	SUD rate given SMI	9.75%	NSDUH 2023 detailed tables
ω_ℓ	average hours worked	40% of time	Guvenen et al. (2023)
b_1	keeps the drug choice interior, so θ_v identifies the expenditure share	—	—

The discount factor and wealth/GDP. β is calibrated so that aggregate wealth is three times GDP, with GDP measured as *all factor income* — labour earnings plus the return on the capital stock — and transfers excluded, since transfers are redistribution rather than output. Three is the standard US ratio of the private capital stock to annual output and is the moment a life-cycle discount factor is normally pinned by.

This is the right target here rather than the SCF's net-worth-to-household-income ratio of roughly 7.3, whose numerator is mostly housing equity that a model without homeownership cannot hold and whose denominator is a household income concept rather than output. One implication should be stated in the paper rather than left for a reader to find: with $A = 3 \cdot \text{GDP}$

and GDP = labour income $+ rA$, capital income is 9% of GDP, which is low against a one-third capital share. That follows from holding r fixed at 3% in a model with no production function.

Three targets that were wrong in the submitted paper. These are worth flagging because two referees used them to distrust the quantification.

1. *SUD among the homeless.* The paper targets 37%. Gutwinski et al. (2021) pool 39 studies of homeless people in high-income countries and report 36.7% *alcohol* use disorder and 21.7% *drug* use disorder. The model is about drug use, so the target is 21.7%. This answers the AE's A16.

2. *Income of drug users.* Referee 1 (2c-i) attacked the model's 41% against "roughly 74%" in the data. The 74% is computed over people with a use *disorder*, not people who use — a different population. For ages 26+, NSDUH gives 95.1% for users and 68.7% for moderate or severe use disorder. Targeting the disorder ratio and *validating* on the use ratio is the sharper test, and it makes the economics explicit: use is close to income-neutral, and it is progression to disorder that concentrates at the bottom of the distribution.

3. *Housing burden.* Everyone rents in this model, so a renters-only target would import the selection into who rents. The target is therefore the aggregate housing-cost-to-income ratio over *all* households, with owner housing valued at imputed rent, and the full distribution is used for validation (Section 3).

2.2 Second stage: short-run experimental moments

Every report converges on the same objection. Referee 2 (C1): "all of the welfare results rest on whether the model has an accurate elasticity of drug use to housing access". Referee 3 and the AE (A13) say the same independently. Referee 1 (2d, 2e) asks for policy-relevant elasticities from the RCT literature and notes that the price-elasticity moment as written is not replicable.

These are short-run objects. A randomised trial hands a treatment to a selected group and measures them months later; it does not move aggregate prices and it does not wait for a new stationary distribution. The second stage therefore holds prices at their benchmark values, takes the benchmark distribution as the initial condition, restricts attention to the trial's eligible population, applies each arm's rules, re-solves the household problem, and pushes the cohort forward.

2.2.1 The housing-access elasticity

Milby et al. (2005) randomised 196 cocaine-dependent homeless adults into three arms, all receiving identical day treatment:

Arm	Meaning here	n	Weeks abstinent (of 24)
NH	no assistance	66	4.48
NACH	housing, unconditional — *Housing First*	67	6.60
ACH	housing conditional on abstinence — *Treatment First*	63	8.32

The model counterparts are the same three arms applied to the model's homeless population with SUD, and the moment targeted is the *ratio* NACH/NH = 1.473. Ratios rather than levels: "mean consecutive weeks abstinent out of 24" is not the same object as "abstains this model year", but the treatment effect is comparable, and it is the treatment effect the referees are asking the model to reproduce. This moment identifies ρ_H in (2).

The conditionality effect is a validation check, not a target. The abstract of Milby et al. reports that the abstinence-contingent arm beat the unconditional arm, but that holds only *after controlling for treatment attendance*. In the raw randomised comparison NACH versus ACH gives $P = .091$, and the GEE models state that the two housed groups "did not differ from one another". Calibrating to a point estimate the source paper reports as insignificant would be exactly the error Referee 1 attacked κ_{cons} for. The model's ACH/NACH ratio is therefore reported against 1.261 as a sign check with the confidence band shown.

The extensive/intensive contrast is the sharper test. Milby's counterfactual is literally no housing, and the effect is large. The Housing First RCTs compare Housing First to "usual services", which already include shelter or transitional housing, and the systematic review finds "no clear differences ... on outcomes related to mental health, quality of life and substance use". These are not in conflict. The elasticity of drug use with respect to housing access is **large at the extensive margin** (nothing to something) and **near zero at the intensive margin** (some housing to permanent housing). A model that reproduces both is far better identified than one that matches either number alone, and reproducing both is the test this paper should set itself.

2.2.2 The drug-price participation elasticity

Referee 1 (2e) says this moment is not replicable as written. It now has an unambiguous model counterpart: perturb p_v, hold everything else at the benchmark, re-solve the household problem, and read the arc elasticity of the participation rate. It is a partial-equilibrium perturbation deliberately, since the empirical estimates come from price variation rather than from a comparison of steady states. The target is -0.63 and it identifies $\kappa_{\phi,d}$, the SUD mortality multiplier, whose admissible range is the 2.5–9.8 standardised-mortality-ratio band of Degenhardt et al. (2011). Referee 1 also asks for robustness across the -0.36 to -0.9 range, which is a re-run of the same bisection at the two endpoints.

2.2.3 The treatment effect and the value of a statistical life

The treatment effect on the recovery hazard, g, is the parameter through which the entire Treatment First result runs, so it deserves the same discipline Referee 1 forced on κ_{cons}. The submitted value of 3.0 — tripling the recovery hazard — is not supported. Remission from SUD overall runs 35–54% at a mean 17-year follow-up and remission following specialty treatment about 48%, but those are different populations with different selection, not treatment effects. The only clean randomised estimate is an odds ratio near 1.22 for contingency management, which implies $g \approx 1.35$. The paper should report the Treatment First result across a range and identify the value of g at which the ranking would change.

The model-implied value of a statistical life is the marginal rate of substitution between consumption and the survival probability. It is currently *reported* rather than targeted, because $\aleph$ carries a first-stage moment; the reference value is \$7.5m (Viscusi and Aldy 2003; EPA 2010). If the model-implied value is far from that, the right response is to move $\aleph$ to the VSL target and reassign the SUD income ratio to $\rho_{\text{pen},d}$.

3 Validation

None of the moments in this section is targeted. Referee 1's comment 2(a) — that model fit is never reported — and the AE's A12 are answered by reporting all of them, including the misses, and by framing the misses honestly rather than favourably (comment 2b).

3.1 Distributional moments

Table 4: Untargeted distributional moments

Moment	Data	Source
Distribution of housing cost / income, 12 thresholds	95.4, 89.3, 77.2, 63.6, 50.9, 40.6, 32.5, 26.0, 17.5, 12.6, 7.1, 4.6% above 5–100%	SCF 2022, ages 26–85, owner housing at imputed rent
Income of drug *users* relative to average	95.1%	NSDUH 2024, ages 26+
Share of the homeless who are employed	40–52%	Referee 1, comment 2(c)(i)
Age distribution of the homeless	48.1% aged 26–44; 27.1% aged 55+	Referee 1, comment 2(c)(ii)
Drug use by age (U-shaped)	NSDUH	microdata
Homelessness by age	California point-in-time counts	administrative
Wealth by age	SCF 2022	microdata

3.2 Policy responses

Table 5: Untargeted policy responses

Response	Data	Source
Housing First effect on 3-year mortality	−4.6pp (s.e. 1.7)	Costantini (2025), ≈300,000 VA veterans, IV
Treatment First effect on long-run health	≈ 0	Costantini (2025)
Housing stability under Housing First	sustained	HF RCT literature

Response	Data	Source
Housing stability under Treatment First	declining	HF RCT literature
Property values: permanent supportive housing versus emergency shelter	PSH raises, shelter lowers	shelter-siting hedonic literature

The Costantini result is the paper's largest risk and belongs in the introduction. The best available quasi-experimental evidence says the mortality benefit runs through *housing*, and that conditioning on treatment adds nothing detectable. The submitted paper concludes that universal Treatment First dominates. The defensible position after the revision is the *generosity* result — expanding coverage dominates, and conditionality is second-order and uncertain — which is consistent with all of the evidence above and is still a paper. Getting there requires reporting welfare and mortality as two separate dimensions, which is the AE's A9, rather than collapsing them into one ranking.

4 Mechanisms: shutting channels and recalibrating b

4.1 Why this replaces the static appendix

Referee 1's comment 1(d) does not merely ask for intuition, it *specifies the design*: show that the mortality channel is the quantitatively important one by shutting channels off. Comment 1(e) says the current appendix is disorganised, static, and overreaches. Referee 2 (C4) wants the drug-use policy function shown by income and wealth for households with and without SUD. Referee 3 (M2b) asks what drug-consumption patterns the Stone–Geary specification actually generates and where heterogeneity in v comes from; (M2c) asks the sharpest theoretical question in any of the reports — since drug use is rational and forward-looking, where do the externalities justifying intervention arise? The AE (A1) asks for a stylised section because the number of moving parts obscures the economics.

One exercise answers all five, and it is run on the *calibrated* model rather than on a separate static toy. That matters: a static model with no assets cannot speak to the question Referee 2 raised about wealth, and a toy calibrated to nothing cannot establish that a channel is quantitatively important.

4.2 The design

Shut one channel off at a time and, in each arm, **recalibrate b_0 so that drug participation returns to its benchmark value**. The recalibration is what makes the comparison informative. Without it, removing the mortality penalty simply raises drug use everywhere, and the level change swamps any compositional change. Holding participation fixed separates *sorting* from *level* and yields two distinct statistics per arm:

Statistic	What it measures
How far b_0 had to move	how much deterrence the channel was providing
The income gradient of use	*whom* the channel deterred

The arms are run in partial equilibrium at benchmark prices. The object of interest is the behavioural mechanism, not equilibrium feedback, and holding prices fixed isolates it — as well as being roughly fifty times cheaper.

4.3 The arms

1. **Benchmark.** All channels on.
2. **No mortality penalty** ($\kappa_{\phi,d} = 1$). SUD no longer raises the death hazard. This is Referee 1's channel of interest and the one the paper claims is central.
3. **No employment penalty** ($\kappa_{ue,d} = 1$). SUD no longer worsens job finding or separation.
4. **No homelessness feedback** ($\kappa_{\phi,H} = \kappa_{ue,H} = \mu_H = 1$). Being homeless no longer raises mortality, worsens employment, or lowers the recovery hazard. This arm isolates the loop that makes homelessness and SUD mutually reinforcing.
5. **All private costs off at once.** Mortality, employment and the homelessness loop are shut simultaneously. Whatever deterrence survives is *resource diversion*: the budget constraint alone. Diversion cannot be switched off directly — drugs are priced inside a Cobb–Douglas bundle, so making them free sends v to the corner — but it is measurable as the residual, which is what this arm reports.

4.4 What is reported

For each arm, Table 6 reports the recalibrated b_0 relative to benchmark, and then the composition of drug users: the use rate within each income quintile, mean user income relative to average (split into experimenters and households with SUD), and the shares of users who are unemployed and homeless, together with their mean age and assets.

4.5 What the exercise is expected to show, and how it could fail

The paper's thesis is that the private cost of SUD runs through mortality, employment and resource diversion, and that this is why forward-looking households with more to lose — higher income, more assets, better health — abstain. Fixing preferences across SUD status (Section 2) makes this a commitment rather than an assumption: with α, β, γ invariant, becoming addicted no longer relaxes any constraint, so *all* of the deterrence must come from those three channels.

The exercise therefore has a sharp prediction and a sharp failure mode. If removing the mortality penalty requires only a small change in b_0 and leaves the income gradient of use nearly flat, the paper's central mechanism is not carrying the weight the text claims, and the honest response is to say so and re-centre the paper on the generosity result. If instead b_0 must fall substantially and the income gradient inverts, the mechanism is established numerically on the

Table 6: Mechanism decomposition (structure; numbers pending calibration)

	Benchmark	No mortality penalty	No employment penalty	No homeless feedback	All private costs off
b_0 relative to benchmark	1.00				
Participation (%, held fixed)					
SUD rate (%)					
Homelessness (%)					
Use rate, income Q1 (%)					
Use rate, income Q5 (%)					
User income / average					
experimenters only					
SUD only					
Users unemployed (%)					
Users homeless (%)					
Mean age of users					
Mean assets of users					

calibrated model — which is exactly what Referee 1 asked for and what the static appendix could not deliver.

4.6 Where the externalities come from

Referee 3's M2c asks where the externalities justifying intervention arise, given that drug use is rational and forward-looking. They are of two kinds and the shutdown ladder quantifies both.

Fiscal. A household that becomes homeless or acquires SUD imposes χ_H and χ_D on the government budget and reduces the tax base through lower earnings. None of this is internalised, and all of it is measured directly by the model.

Non-fiscal. The disamenity term $\kappa_x(H + \omega_x H_D)$ in (1). Referee 2 (C2) asked for this and said it was easy to introduce; it is, but it has no credible point estimate. Hedonic shelter-siting studies measure a *local, avoidable* disamenity — what people pay to live further away — and this model has no geography, so nobody can avoid anything and a capitalisation coefficient does not map onto an aggregate public bad. Corbae, Glover and Nattinger's externality is about neighbourhood housing *quality* rather than homelessness, and their identification comes from a rent-decontrol experiment, so what transfers is their identification logic and not their functional form.

The recommendation is therefore to set $\kappa_x = 0$ in the baseline, report a bounded sensitivity, and use the qualitative finding from that literature — that permanent supportive housing *raises* nearby property values while temporary emergency shelter lowers them — as the substantive point. Permanent supportive housing is Housing First-like and transitional shelter is Treatment First-like, so the externality plausibly differs in *sign* between the two policies being compared. That is a stronger argument than a made-up parameter, and note that $\kappa_x = 0$ is a defensible baseline in a way that $\kappa_{\text{cons}} = 0$ was not: it was the value at which the paper's conclusion reversed.

5 Status

Done. Household problem with continuous labour supply from the first-order condition; stationary distribution; aggregates; the three equilibrium fixed points; the calibration framework with all nineteen parameter–moment assignments; the second-stage experiment machinery; the mechanism ladder.

Running. The first-stage calibration. Every number the model currently prints is a placeholder and none should be read as a result.

Not yet built. The transition-path solver. Referee 1's comment 3(d) calls the several-period partial-equilibrium exercise "a half-measure" and asks for proper transition paths; with no housing market to clear, a path iterates on τ_ℓ and p_v alone, which is far cheaper than in the original model. This is the critical path for the revision, because the second-stage moments of Section 2.2 are short-run objects.

Open decisions. Whether to add a parental warm-glow term to answer Referee 2's C3 and the AE's A6 together, or simply to delete the intergenerational language; and whether to retain the maximum experimentation age, which Referee 1 correctly notes is in tension with the U-shaped age profile of drug use.